

# THE KB-DUAL HYPOTHESIS IS NECESSARY FOR D-DOMINATION

AGENT LANE 04

STATUS

This packet gives a full answer to the necessity question following Theorem 3 in Emelyanov–Erkursun-Ozcan–Gorokhova, *d-Operators in Banach Lattices* [1]. Their theorem proves that if  $E'$  is a KB-space, then domination by a positive d-compact, respectively d-weakly compact, operator passes to positive dominated operators. They ask whether the condition on  $E'$  is necessary. It is necessary for both conclusions, in the natural uniform sense over all Banach lattice ranges.

SOURCE QUESTION

After proving the domination theorem, the source states:

We do not know whether or not the condition on  $E'$  in Theorem 3 is necessary for conclusions in i) and/or in ii).

By the Wickstead theorem, (cf., [2, Theorem 5.31]), if either  $E'$  or  $F'$  has order continuous norm then  $0 \leq S \leq T \in W(E, F)$  implies  $S \in W(E, F)$ . The following theorem deal with the domination problem for d-(weakly) compact operators.

**Theorem 3.** Let  $0 \leq S \leq T \in L(E, F)$ , where  $E'$  is a KB-space.

i) If  $T \in \text{d-K}(E, F)$  then  $S \in \text{d-K}(E, F)$ .

ii) If  $T \in \text{d-W}(E, F)$  then  $S \in \text{d-W}(E, F)$ .

*Proof.* i) Let  $D$  be a disjoint normalized countable subset of  $E_+$ . Accordingly to Proposition 2, we need to show that  $S(D)$  is relatively compact. So, let  $(y_n)$  be a sequence in  $S(D)$ . If  $y_n = 0$  for infinitely many  $n$ -th then  $(y_n)$  has a norm-null subsequence. If  $y_n = 0$  only for finitely many  $n$ -th, by removing them, we may suppose all  $y_n \neq 0$ . For each  $n$ , pick  $x_n \in D$  with  $y_n = Sx_n$  and denote  $D_0 = \{x_n : n \in \mathbb{N}\}$ . Then  $D_0$  is a disjoint normalized subset of  $E_+$ . If  $D_0$  is finite then  $S(D_0)$  is finite and hence compact. So, suppose  $D_0$  is infinite. By the assumption i),  $T(D_0)$  is relatively compact. Since  $E'$  is a KB-space, the disjoint normalized sequence  $(x_n)$  is w-null, and hence  $Tx_n \xrightarrow{w} 0$ . Suppose  $\|Tx_n\| \not\xrightarrow{w} 0$ . Then  $\|Tx_{n_k}\| \geq M > 0$  for some subsequence  $(x_{n_k})$  of  $(x_n)$ . Since  $T \in \text{d-K}(E, F)$ , there exists a further subsequence  $(x_{n_{k_j}})$  such that  $\|Tx_{n_{k_j}} - z\| \rightarrow 0$ , and hence  $Tx_{n_{k_j}} \xrightarrow{w} z$ . Since  $Tx_n \xrightarrow{w} 0$  then  $z = 0$ , and hence  $\|Tx_{n_{k_j}}\| \rightarrow 0$  violating  $\|Tx_{n_k}\| \geq M > 0$  for all  $k$ . The obtained contradiction proved  $\|Tx_n\| \rightarrow 0$ . Using  $0 \leq S \leq T$ , we obtain  $\|y_n\| = \|Sx_n\| \rightarrow 0$ . Thus  $S \in \text{d-K}(E, F)$ .

ii) It follows from Theorem 2. □

We do not know whether or not the condition on  $E'$  in Theorem 3 is necessary for conclusions in i) and/or in ii).

## DEFINITIONS

Let  $E$  and  $F$  be Banach lattices. A bounded operator  $T : E \rightarrow F$  is d-compact, written  $T \in \text{d-K}(E, F)$ , if  $T(D)$  is relatively norm compact for every disjoint bounded set  $D \subset E$ . It is d-weakly compact, written  $T \in \text{d-W}(E, F)$ , if  $T(D)$  is relatively weakly compact for every disjoint bounded set  $D \subset E$ .

## IDEA OF THE PROOF

If  $E'$  is not a KB-space, then  $E$  has a bounded positive disjoint sequence  $(x_n)$  which is not weakly null. A positive functional  $u \in E'_+$  sees a subsequence uniformly:  $u(x_n) \geq \delta > 0$ . Split  $u$  into disjoint positive pieces  $\varphi_n$  supported on the disjoint carriers of the  $x_n$ 's. The rank-one operator

$$Tx = u(x)\mathbf{1} \quad (E \rightarrow c)$$

is compact. The dominated positive operator

$$Sx = \left( \sum_{i \geq k} \varphi_i(x) \right)_{k=1}^\infty$$

sends  $x_n$  to an initial-segment vector of height  $u(x_n)$ . These initial-segment vectors have no weakly convergent subsequence in  $c$ , because coordinate functionals force any weak limit to have all coordinates equal to a positive number, while the limit functional on  $c$  forces the limit at infinity to be zero. Hence  $S$  is not d-weakly compact, and therefore not d-compact.

**Lemma 1** (disjoint splitting of a positive functional). *Let  $E$  be a Banach lattice, let  $u \in E'_+$ , and let  $(x_n)$  be a disjoint sequence in  $E_+$ . There are pairwise disjoint functionals  $\varphi_n \in E'_+$  such that  $0 \leq \varphi_n \leq u$ ,*

$$\sum_{n \in A} \varphi_n \leq u \quad \text{for every finite } A \subset \mathbb{N},$$

and

$$\varphi_n(x_m) = \begin{cases} u(x_n), & m = n, \\ 0, & m \neq n. \end{cases}$$

*Proof.* This is the standard carrier-projection decomposition in the order dual. The norm dual  $E'$  is Dedekind complete as a vector lattice. For each  $n$ , let  $P_n$  be the band projection in  $E'$  onto the carrier band associated with the principal band generated by  $x_n$ . Since the vectors  $x_n$  are disjoint, these carrier bands are pairwise disjoint. Put  $\varphi_n = P_n u$ . Then the  $\varphi_n$  are pairwise disjoint positive components of  $u$ , so finite sums are dominated by  $u$ . The component  $u - P_n u$  vanishes on the principal band of  $x_n$ , while  $P_n u$  vanishes on the disjoint principal bands generated by  $x_m$  for  $m \neq n$ , giving the displayed identities.  $\square$

## MAIN THEOREM

**Theorem 1.** *For a Banach lattice  $E$ , the following are equivalent.*

- (a)  $E'$  is a KB-space.
- (b) For every Banach lattice  $F$  and all  $0 \leq S \leq T$  in  $\mathcal{L}(E, F)$ ,  $T \in \text{d-K}(E, F)$  implies  $S \in \text{d-K}(E, F)$ .
- (c) For every Banach lattice  $F$  and all  $0 \leq S \leq T$  in  $\mathcal{L}(E, F)$ ,  $T \in \text{d-W}(E, F)$  implies  $S \in \text{d-W}(E, F)$ .

*Proof.* The implications (a) $\Rightarrow$ (b) and (a) $\Rightarrow$ (c) are exactly Theorem 3 of [1].

We prove the converse by showing that if  $E'$  is not a KB-space, then both domination principles fail with the same target lattice  $F = c$ . By the standard characterization of KB-duals used in [2, Theorem 2.4.14], there is a bounded disjoint sequence  $(x_n)$  in  $E_+$  which is not weakly null. Passing to a subsequence, choose  $u \in E'_+$  and  $\delta > 0$  such that  $u(x_n) \geq \delta$  for all  $n$ .

Apply Lemma 1 to obtain pairwise disjoint  $\varphi_n \in E'_+$  with finite sums dominated by  $u$  and  $\varphi_n(x_m) = 0$  for  $m \neq n$ ,  $\varphi_n(x_n) = u(x_n)$ . For  $k \geq 1$ , define

$$g_k = \sup_{N \geq k} \sum_{i=k}^N \varphi_i = \sum_{i \geq k} \varphi_i \in E'_+.$$

The order sum exists in the Dedekind complete lattice  $E'$  and satisfies  $0 \leq g_k \leq u$  and  $g_k \downarrow 0$ . In particular,  $g_k(x) \rightarrow 0$  for every  $x \in E$ .

Define positive operators  $S : E \rightarrow c$  and  $T : E \rightarrow c$  by

$$Sx = (g_k(x))_{k=1}^\infty, \quad Tx = u(x)\mathbf{1},$$

where  $\mathbf{1} = (1, 1, \dots)$ . Since  $\|g_k\| \leq \|u\|$  for all  $k$ , the operator  $S$  is bounded. The operator  $T$  is rank one, hence compact and in particular d-compact and d-weakly compact. Also, for  $x \geq 0$ ,

$$0 \leq g_k(x) \leq u(x) \quad (k \geq 1),$$

so  $0 \leq S \leq T$ .

It remains to see that  $S$  is not d-weakly compact. By boundedness of  $(u(x_n))$ , pass to a further subsequence, still denoted  $(x_n)$ , such that  $u(x_n) \rightarrow a$  for some  $a \geq \delta > 0$ . For each  $n$ ,

$$Sx_n = u(x_n)(\underbrace{1, \dots, 1}_{n \text{ times}}, 0, 0, \dots).$$

Let  $(Sx_{n_j})$  be any subsequence. If it were weakly convergent in  $c$  to some  $y \in c$ , then each coordinate functional would give  $y_k = a$  for every fixed  $k$ . Hence  $y$  would have coordinate sequence constantly equal to  $a$ , so its limit at infinity would be  $a$ . On the other hand, the limit functional  $\ell_\infty : c \rightarrow \mathbb{R}$ ,  $\ell_\infty((t_k)) = \lim_k t_k$ , satisfies  $\ell_\infty(Sx_{n_j}) = 0$  for every  $j$ , and weak convergence would force  $\ell_\infty(y) = 0$ . This contradiction shows that  $\{Sx_n : n \in \mathbb{N}\}$  is not relatively weakly compact.

The set  $\{x_n : n \in \mathbb{N}\}$  is disjoint and bounded, so  $S \notin \text{d-W}(E, c)$ . Since every relatively norm compact set is relatively weakly compact,  $S$  is not d-compact either. Thus both (b) and (c) fail whenever  $E'$  is not a KB-space.  $\square$

#### NOVELTY AND VERIFICATION NOTES

- Local indexes were searched for 2401.08792, the source title, and the core phrases **d-compact**, **d-weakly compact**, **domination**, and **KB-space**. No duplicate packet or attempt for this exact question was found.
- Bounded web search on 2026-06-29 for the source title and phrases such as “d-weakly compact domination KB-space” surfaced the source arXiv page but no later resolution.
- The proof deliberately uses  $c$  rather than  $c_0$  so that the rank-one constant operator  $Tx = u(x)\mathbf{1}$  dominates  $S$ .
- The forward implication is not reproved; it is the source Theorem 3. The new content is the converse construction from a non-KB dual.

## HUMAN REVIEW RECOMMENDATION

Review as a likely-valid full answer to the necessity question after Theorem 3 of [1]. The only nontrivial standard ingredient to check is Lemma 1, the carrier decomposition of a positive functional along a disjoint positive sequence. Once that is accepted, the domination counterexample is explicit.

## REFERENCES

- [1] E. Emelyanov, N. Erkursun-Ozcan, and S. Gorokhova, *d-Operators in Banach Lattices*, arXiv:2401.08792, 2024.  
<https://arxiv.org/abs/2401.08792>
- [2] P. Meyer-Nieberg, *Banach Lattices*, Universitext, Springer, 1991.